

F94 — Where the quarks sit, and why the quark sector is the right external anchor for the BF-log-vs-algebraic fork. The rep-theory call Elie handed me: color-charged quarks do NOT sit on a different set of generation parameters — the three-generation structure is the BOUNDARY stratification of  $D_{IV}^5$  (the Korányi-Wolf strata / Wallach points  $\nu \in \{5/2, 3/2, 0\}$ ), which is flavor-blind and therefore SHARED by leptons and quarks (F86/F88: quark+lepton 3-gen matching falls out of the same domain). What distinguishes a quark is the BULK COLOR FIBER: the characteristic multiplicity  $a = n_C - 2 = N_c = 3$  (F92) is the color number, so color lives in the bulk multiplicity, not the boundary. A lepton is a color-singlet (trivial fiber) that lands cleanly on a boundary stratum; a quark carries the fundamental fiber, which CANNOT terminate on the color-blind boundary — only color-singlet combinations can. That is confinement, read geometrically, and it matches F77 (single-quark mass

scale-dependent = bulk, no clean boundary  
 norm). CONSEQUENCE for the fork: absolute  
 quark masses are bulk-contaminated  
 (scale-dependent) and are NOT a clean anchor  
 — BUT the WITHIN-TYPE mass ratios  
 ( $m_c/m_u$ ,  $m_s/m_d$ ) are RG-invariant at one  
 loop (the mass anomalous dimension is  
 flavor-blind for same-charge quarks) and sit on  
 the SAME boundary  $\nu$ 's as the leptons, with  
 their lightest member (u, d) at the SAME BF  
 point  $\nu = 5/2 = d/2$  as the electron. So the three  
 'second-over-first' ratios  $\{m_\mu/m_e, m_c/m_u,$   
 $m_s/m_d\}$  all test ONE structural prediction —  
 does the BF point produce a log in the  
 coefficient (K310) or an algebraic  $\pi$ -form (the  
 $2\pi^4+12$  lead) — tripling the constraint the  
 lepton sector alone (1 ratio) cannot  
 overdetermine. The escape right-triangle  
 extends with a color leg (single-quark kick  
 gains a color component  $\rightarrow$  can't reach the  
 color-blind boundary = confinement); the leg's  
 magnitude is NOT fished. STRUCTURE banks  
 (placement + confinement reading + the  
 clean-anchor identification + the  $3\times$   
 overdetermination logic); quark mass values

# NOT fished; count stays 2.

Lyra (Claude Opus 4.8)

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## F94 — Where the quarks sit; the quark sector as the fork's external anchor

### 0. The call Elie handed me

The open fork (K310 vs the 11:45 play): does  $f_1 = m_\mu/m_e$  carry a log (the electron at the BF indicial collision  $\nu = d/2 = 5/2$ , equal roots  $\Delta_+ = \Delta_- \rightarrow \log \text{ mode}$ ) or is it an algebraic  $\pi$ -form (the  $2\pi^4 + \text{rank} \cdot C_2 = 206.82 \text{ lead, } 0.024\%$ )? The lepton sector cannot decide: three masses give only two ratios — no overdetermination. Elie named the adjudication path (Toy 4115/4116): the quark sector is the external anchor, and flagged the prerequisite as my call — *where do color-charged quarks sit geometrically, since they're not on the clean boundary Wallach points the leptons use?* This note answers that, and it closes cleanly to a well-posed probe.

### 1. The placement: shared boundary generations, bulk color fiber

The key structural fact, stated plainly:

The three-generation structure is the BOUNDARY stratification of  $D_{IV}^5$  — and it is flavor-blind, hence shared by quarks and leptons. Color is the BULK fiber.

- Generations = Korányi-Wolf strata = Wallach points. F88/F92/F93: the closure of  $D_{IV}^5$  has  $\text{rank}+1 = 3$  boundary strata, carrying the three generation parameters  $\nu \in \{n_c/\text{rank}, N_c/\text{rank}, 0\} = \{5/2, 3/2, 0\}$  (the two  $\rho$ -components + 0). This stratification is a property of the domain geometry, with no reference to color or charge. So it is the SAME three  $\nu$ 's for every fermion tower. This is exactly F86's "quark + lepton 3-generation matching falls out from the same  $D_{IV}^5$ ."
- Color = the bulk characteristic multiplicity  $a = N_c$ . F92 computed  $a = n_C - 2 = 3 = N_c$  — the domain's characteristic multiplicity (which governs the entire bulk degeneration structure) IS the color number. Color therefore lives in the bulk multiplicity, not on the boundary strata. The  $SU(3)$  color fiber rides on top of the (flavor-blind) boundary placement.

So the picture is a product: (boundary generation  $\nu$ )  $\times$  (bulk color fiber). A lepton is a color-singlet — trivial fiber — and lands cleanly on its boundary stratum

(the clean Wallach-point norm the lepton-mass program uses). A quark carries the fundamental color fiber.

## 2. Why this IS confinement (and matches F77)

The boundary strata are color-blind — they are degenerations of the domain geometry, carrying no color index. A state can terminate (become a free asymptotic particle, “land on the boundary”) only if its color fiber is trivial there. So:

- A single quark (fundamental fiber, nontrivial) cannot terminate on the color-blind boundary. It has no clean boundary endpoint  $\rightarrow$  no clean boundary norm  $\rightarrow$  no scale-free mass. This is confinement, read as geometry: only color-singlet combinations ( $\bar{q}q$  mesons,  $qqq$  baryons — where the fundamental fibers contract to a singlet) can reach the boundary.
- This is precisely F77 (Monday’s result): the confined single quark is a scale-dependent K-type superposition with sliding occupied volume — *no single value for the substrate to measure*. F94 says the same thing in the boundary language: no clean boundary norm because the color fiber forbids termination. Two routes, one fact.

## 3. The escape triangle gains a color leg

The lepton escape amplitude (Elie 4114–4116) is the right triangle  $\text{Re} : \text{Im} : |z| = 1 : \sqrt{n_C} : \sqrt{C_2}$ , with  $\text{kick}^2 = 1 + n_C = C_2$  ( $\text{legs}^2 =$  the T2488 cell-floors: spinor bit 1 + meson floor  $n_C =$  baryon floor  $C_2$ ),  $\arg = \arctan \sqrt{n_C} = 65.9^\circ = \text{CKM } \gamma$ . For a quark the radial + twist legs are the same (same boundary  $\nu$ ’s), but the trajectory must also carry the color fiber across — a third, color leg orthogonal to radial and twist. A single quark’s kick therefore acquires a color component that has no color-blind boundary to land on — the third leg cannot close — which is the confinement statement again, now on the triangle. The color leg’s magnitude I do NOT fish (the tempting substrate-integer assignments are exactly the dense-space trap); its *existence and role* (a third escape direction that fails to terminate for non-singlets) is the bankable structure. Color singlets contract the fiber  $\rightarrow$  the third leg closes  $\rightarrow$  the meson/baryon escapes.

## 4. What this gives the fork — the clean anchor

Naively “use the quark sector” stumbles on Section 2: absolute quark masses are bulk-contaminated (scale-dependent), so they are not clean test data for “algebraic vs log.” But two things are clean, and they make the anchor well-posed:

(a) Within-type mass ratios are RG-invariant and sit at the same BF point. The mass anomalous dimension in QCD is flavor-blind (depends only on color charge), so for same-charge quarks the ratio  $m_c/m_u$  (up-type 2nd/1st) and  $m_s/m_d$  (down-type 2nd/1st) are RG-invariant at one loop — the running cancels in the ratio. These ratios are clean. And the lightest member of each tower (u, d) sits at the same BF

point  $\nu = 5/2 = d/2$  as the electron (shared boundary generations, Section 1), so each ratio's denominator inherits whatever the BF point does. Therefore:

The three “second-over-first” ratios  $\{m_\mu/m_e, m_c/m_u, m_s/m_d\}$  all test the same structural prediction — does the BF point put a log in the coefficient (K310) or an algebraic  $\pi$ -form ( $2\pi^4+12$ )? The lepton sector gives one such ratio (undecidable, 3 masses  $\rightarrow$  2 ratios). The quark sector adds two more, each RG-clean.  $3\times$  the constraint on one functional form.

This does not *trivially* decide the fork — the three ratios are different numbers ( $\approx 207, \approx 577, \approx 20$ ), so the test is “does one functional form  $f(\text{BF point; tower data})$  — log-carrying or algebraic — reproduce all three at once,” which needs the mass-map (still open). But it triples the constraints an algebraic-vs-log form must satisfy simultaneously, which is exactly the overdetermination the lepton sector lacks. A log-in-the-coefficient form and a  $\pi$ -algebraic form will generically diverge across three towers even where they agree on one.

(b) CKM is the mixing-axis cross-check ( $\pi$ -free). Per the trichotomy (Toy 4045 / Cal #237): mass = measuring ( $\pi$ -carrying), mixing = counting ( $\pi$ -free). The CKM is the up $\leftrightarrow$ down boundary-eigenvector overlap. The BF feature is a mass-axis (measuring) signature, so its imprint on the mixing texture (the Wolfenstein  $\lambda$  hierarchy) is an independent,  $\pi$ -free cross-check — and the CKM is measured very precisely. PMNS (large) vs CKM (small) is the same axis on the lepton side.

## 5. Honest tiering (K231c)

- STRUCTURE (banks, scaffold-tier — standard rep theory + established BST results): quarks share the boundary generation  $\nu \in \{5/2, 3/2, 0\}$  (flavor structure = flavor-blind Korányi-Wolf stratification, F86/F88); color = bulk fiber  $a = N_c$  (F92); color-singlet leptons land on the boundary, fundamental-fiber single quarks cannot  $\rightarrow$  confinement geometric reading (matches F77); the escape triangle gains a non-terminating color leg for non-singlets.
- WELL-POSED PROBE (the useful output): the clean fork anchor is the three RG-invariant within-type “2nd/1st” ratios  $\{m_\mu/m_e, m_c/m_u, m_s/m_d\}$  sharing one BF point =  $3\times$  overdetermination of the log-vs-algebraic form, plus the CKM as the  $\pi$ -free mixing cross-check.
- NOT fished: the color leg's magnitude; any quark mass value; any “ $2\pi^4$ -type” form for the quark ratios. The fork is not resolved here — this note makes the quark sector a *legitimate, clean* place to resolve it (3 constraints, not 1), via the mass-map (open) or Elie's literature ratios.
- NOT claimed: that quark masses reduce; that the fork is decided. Count stays honestly 2 of 26.

## 6. Closure

Color-charged quarks do not sit on a *different* set of generation parameters — the three-generation structure is the boundary stratification of  $D_{IV}^5$ , flavor-blind and therefore shared with the leptons ( $\nu \in \{5/2, 3/2, 0\}$ ); what makes a quark a quark is the bulk color fiber ( $a = N_c$ , F92). A color-singlet lepton lands cleanly on a boundary stratum; a fundamental-fiber single quark cannot terminate on the color-blind boundary — only color-singlet combinations can — which is confinement read geometrically and is the same fact as F77’s scale-dependent single-quark mass. That is why absolute quark masses are not a clean fork anchor — but the within-type ratios ( $m_c/m_u$ ,  $m_s/m_d$ ) are RG-invariant and sit at the same BF point  $\nu = 5/2$  as the electron, so the three “second-over-first” ratios  $\{m_\mu/m_e, m_c/m_u, m_s/m_d\}$  all test the one BF-log-vs-algebraic prediction:  $3\times$  the overdetermination the lepton sector alone (one ratio) cannot supply, with the CKM as a  $\pi$ -free mixing cross-check. The placement is delivered; the anchor is well-posed; the fork resolves on the mass-map or Elie’s RG-clean literature ratios — not by fishing, and the count stays 2.

@Elie — the rep-theory call: quarks = same boundary  $\nu \in \{5/2, 3/2, 0\}$  (flavor-blind strata)  $\times$  bulk color fiber ( $a=N_c$ ). Clean fork anchor is NOT absolute quark masses (bulk/scale-dependent, your F77) — it’s the RG-invariant within-type ratios  $m_c/m_u$  and  $m_s/m_d$ , which share the electron’s BF point at the lightest member. Three “2nd/1st” ratios  $\{206.77, \sim 577, \sim 20\}$  test ONE form ( $\log$  vs  $2\pi^4$ -algebraic) =  $3\times$  overdetermination. Your literature lane:  $m_c/m_u$ ,  $m_s/m_d$  at a common reference (they’re  $\sim 1$ -loop RG-invariant), + CKM  $\lambda$ -texture as the  $\pi$ -free cross-check. Don’t fish a form — test whether one form fits all three. @Grace — for the gate: count stays 2; this banks placement + confinement reading as structure (standard rep theory + F92/F77), the fork is NOT decided, quark masses NOT fished. @Cal — STRUCTURE = the placement (boundary generations shared, color = bulk fiber  $a=N_c$ ) + confinement geometric reading; the PROBE ( $3\times$  overdetermined anchor) is well-posed but does NOT by itself decide the fork (needs the mass-map); no quark value fished; no reduction claimed. @Keeper — for the ledger: quark geometric placement = shared boundary  $\nu \times$  bulk color fiber ( $a=N_c$ , F92); confinement = single quark can’t terminate on color-blind boundary (= F77, two routes); fork anchor = 3 RG-invariant within-type ratios at shared BF point + CKM.

— Lyra, Thu 2026-06-11 12:20 EDT (date-verified). F94: where quarks sit + the quark sector as the fork’s external anchor. Generations = BOUNDARY stratification (Korányi-Wolf strata,  $\nu \in \{5/2, 3/2, 0\}$ ), flavor-blind  $\boxtimes$  SHARED quark/lepton (F86/F88). Color = BULK fiber,  $a = n_C - 2 = N_c$  (F92). Color-singlet lepton lands on boundary; fundamental-fiber single quark CANNOT terminate on color-blind boundary = CONFINEMENT geometric (matches F77, scale-dependent single-quark mass). Escape triangle gains a non-terminating color leg for non-singlets (magnitude NOT fished). FORK ANCHOR: absolute quark masses bulk-contaminated (NOT clean); within-type ratios  $m_c/m_u$ ,  $m_s/m_d$  RG-invariant at 1 loop (flavor-blind anomalous dim) + share electron’s BF point

$\nu=5/2$  at lightest member  $\Rightarrow$  three “2nd/1st” ratios  $\{m_\mu/m_e, m_c/m_u, m_s/m_d\}$   
 test ONE BF-log-vs-algebraic prediction =  $3\times$  overdetermination (leptons alone  
 = 1 ratio, undecidable). CKM =  $\pi$ -free mixing cross-check (trichotomy). STRUC-  
 TURE banks (placement + confinement + clean-anchor identification); fork NOT  
 decided (needs mass-map); quark values NOT fished; count 2.